Stakeholder memo - Travel Booking BI

For: Marketing, Operations, Finance | Data: 119,390 bookings, Jul 2015 - Aug 2017 (clean fact 119,209)

What we found

1. Volume is not margin. TA/TO delivers 82% of bookings but cancels at 41.1% and returns ROI 3.15 after channel costs. Direct is 12% of bookings, cancels at 17.5%, and returns ROI 20.40 with 95.3% gross margin. Shifting mix toward Direct is worth more than buying more TA/TO volume.

2. Long-lead bookings are inventory risk. Cancellation climbs from 9.6% for arrivals booked within a week to 67.7% beyond a year. Treat bookings with lead time over 180 days as provisional in every forecast and overbooking decision.

3. Revenue run rate: the 13-week Holt-Winters forecast is EUR 3,971,283. Quote it with its holdout error (MAE EUR 19,101 per week) - it beat seasonal naive (48,154) and a 4-week average (97,405) on the same holdout.

4. Flexible free-cancellation rate (simulated A/B test): conversion +9.7% ($p = 0.013$), SRM clean. Cancellation guardrail moved the wrong way but is not significant ($p = 0.182$) and the test is underpowered for it. Recommendation: ship with 4 weeks of weekly cancellation monitoring and a pre-set stop rule.

What we are NOT claiming

- 'Non Refund' deposit bookings show 99.4% canceled in the source data. That is a data quirk, not a finding. No deposit policy should change until Finance verifies how non-refundable revenue is recorded.
- Margin and ROI use stated, synthetic channel-cost rates (reference file in the repo). The Direct vs TA/TO ranking is robust to those rates; the exact ROI values are not.

Ask

Marketing: agree a Direct-share target for next quarter. Operations: adopt the lead-time risk bands in the Tableau Operations page. Finance: verify the Non Refund recording question so the deposit view can be trusted.